Build Solution

Given a use case, identify the considerations when choosing the right option in making an outbound call to an external system.

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After studying this topic, you should be able to:

- List the outbound integration options offered by Salesforce.
- Identify the considerations for using External Services.
- Identify the considerations for making REST and SOAP callouts from Apex code.
- Identify the considerations for sending an outbound message to an external system.

Introduction

Salesforce offers various options for making outbound calls from Salesforce. These include **External Services**, **Apex SOAP Callouts**, **Apex REST Callouts**, and **Outbound Messaging**.

Several considerations apply when utilizing an outbound integration technique. For example, a **remote site setting** must be created to make a REST or SOAP callout using Apex code. A **Named Credential** must be defined for invoking an external service action from a flow. **Outbound Messaging** can be configured to send a session ID to allow the external system to make callbacks.

Overview

External Services

External Services can be used to invoke an external RESTful service if its API specification is available in OpenAPI 2.0 or OpenAPI 3.0 JSON schema format.

REST Callouts

Apex code can be used to perform a REST/HTTP callout to an external system. A remote site setting must be created to authorize the endpoint. HTTP methods such as GET, POST, and PUT can be used.

SOAP Callouts

Apex code can be used to perform a SOAP callout to an external system. A remote site setting must be created in Salesforce to authorize the endpoint. The WSDL of the web service must be consumed to generate Apex classes.

Outbound Messaging

A flow can be used to send an outbound message to a SOAP endpoint. It can be sent as a specific user. The session ID can be included in the message. Its WSDL can be used to create a listener.

Outbound Integration

Various options are available for making **outbound calls** from Salesforce to an external system.

EXTERNAL SERVICES

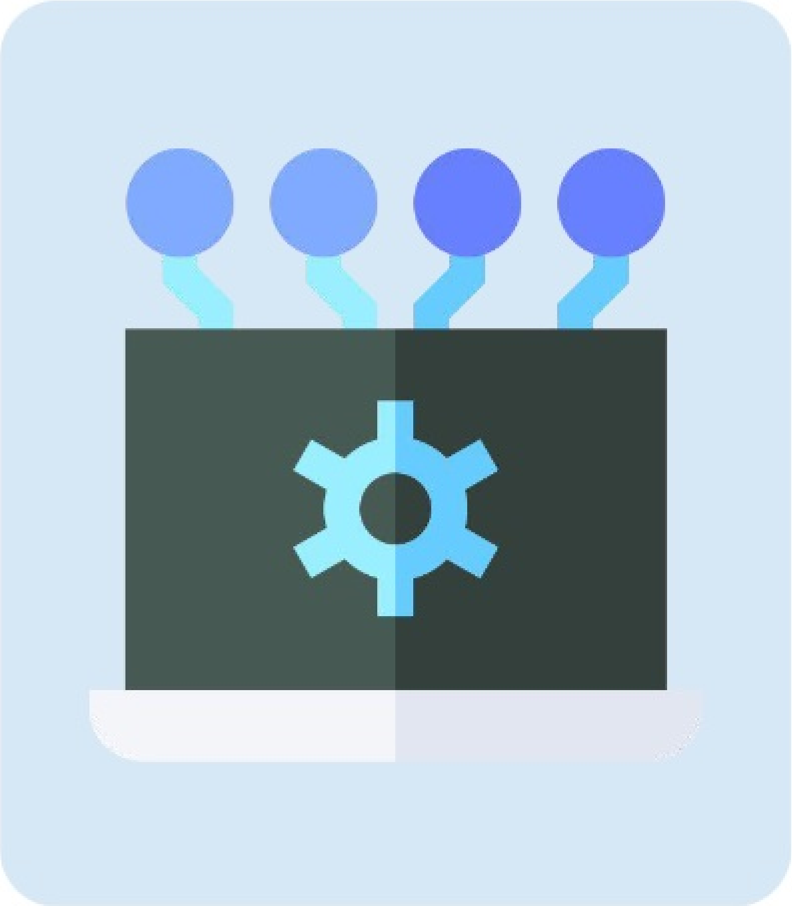
External Services can be used to invoke an external service through a **REST API call**.

REST/SOAP CALLOUT

A **synchronous or asynchronous Apex REST or SOAP callout** to an external system can be invoked from a **Lightning component, Lightning page, Visualforce page, button, or batch Apex job**. An **Apex trigger** can perform an **asynchronous Apex REST or SOAP callout**.

OUTBOUND MESSAGING

A **flow** can be used to send an **outbound message** to a SOAP endpoint asynchronously.



External Services

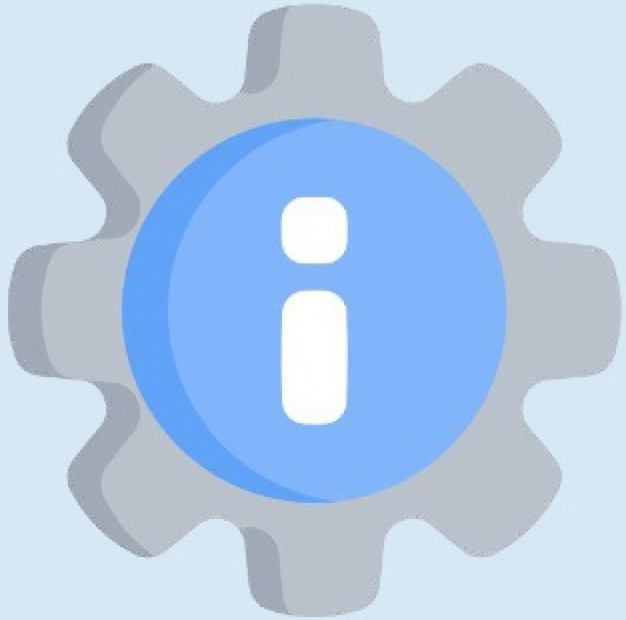
There are various considerations for using External Services to **invoke an externally hosted service** in a declarative or programmatic manner.

PREREQUISITES

The external service must be a **RESTful service**, and its **API specification** must be available in **OpenAPI 2.0** or **OpenAPI 3.0 JSON schema** format. It should be possible to invoke the transactions from **Apex, Flow, or Einstein bots**.

AUTHENTICATION

A **Named Credential** must be defined to use External Services. Storing authentication data in a flow is **not supported**. Additionally, defining a schema's **securityDefinitions** and **security** sections is not supported.



External Services

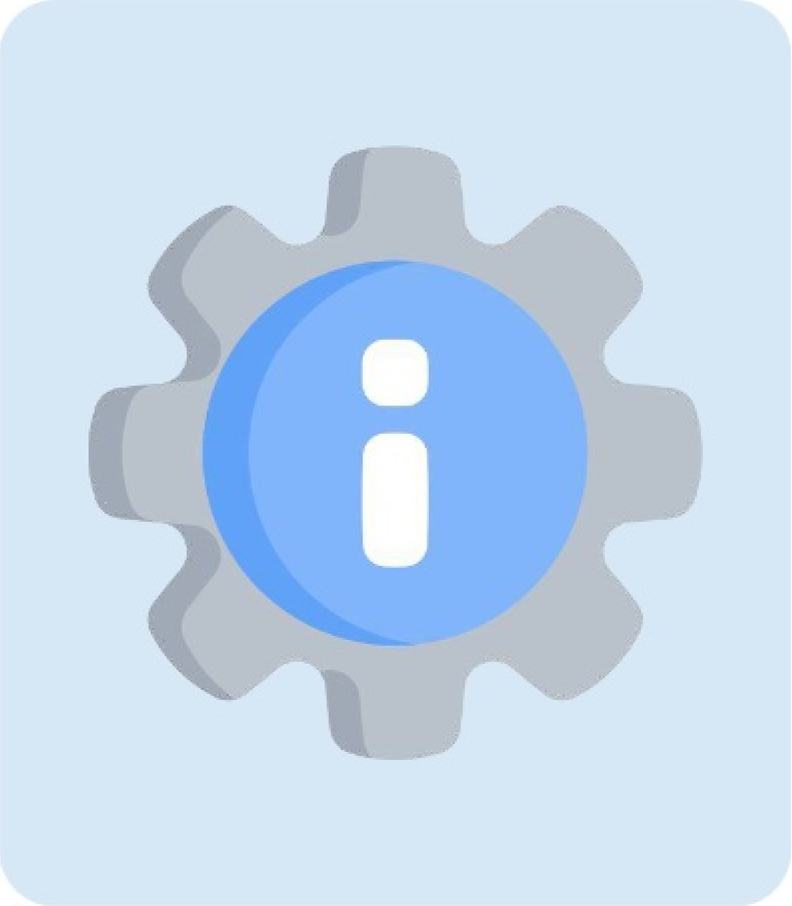
There are various considerations for using External Services to **invoke an externally hosted service** in a declarative or programmatic manner.

REGISTRATION

The External Services wizard can be used to register the external web service. The **API spec** and **Named Credential** are used during the registration process.

SCHEMA

Salesforce provides several options for specifying the **API spec** that will be used by an external service: providing a **relative** or **absolute URL**, **copy-pasting** the **schema text**, or **uploading** the **schema file** in JSON format. During registration, the schema will be automatically **validated**.



External Services

There are various considerations for using External Services to **invoke an externally hosted service** in a declarative or programmatic manner.

APEX CLASSES

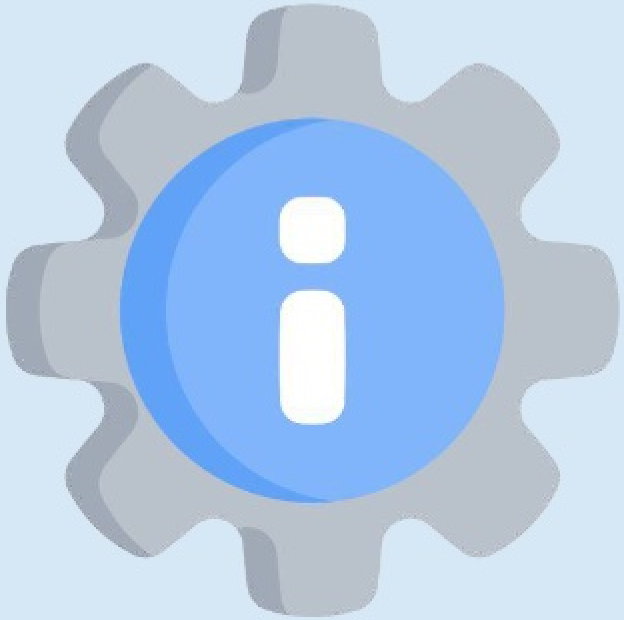
All auto-generated Apex classes in External Services can be accessed in the **Apex Classes viewer** in Setup.

EXTERNAL SERVICES ACTIONS

All the **operations** defined in the **API spec** are available for use within a **flow** as matching actions.

APEX CALLOUTS

Apex code can be used to create a **callout** to the **registered external service**.



External Services

There are various considerations for using External Services to **invoke an externally hosted service** in a declarative or programmatic manner.

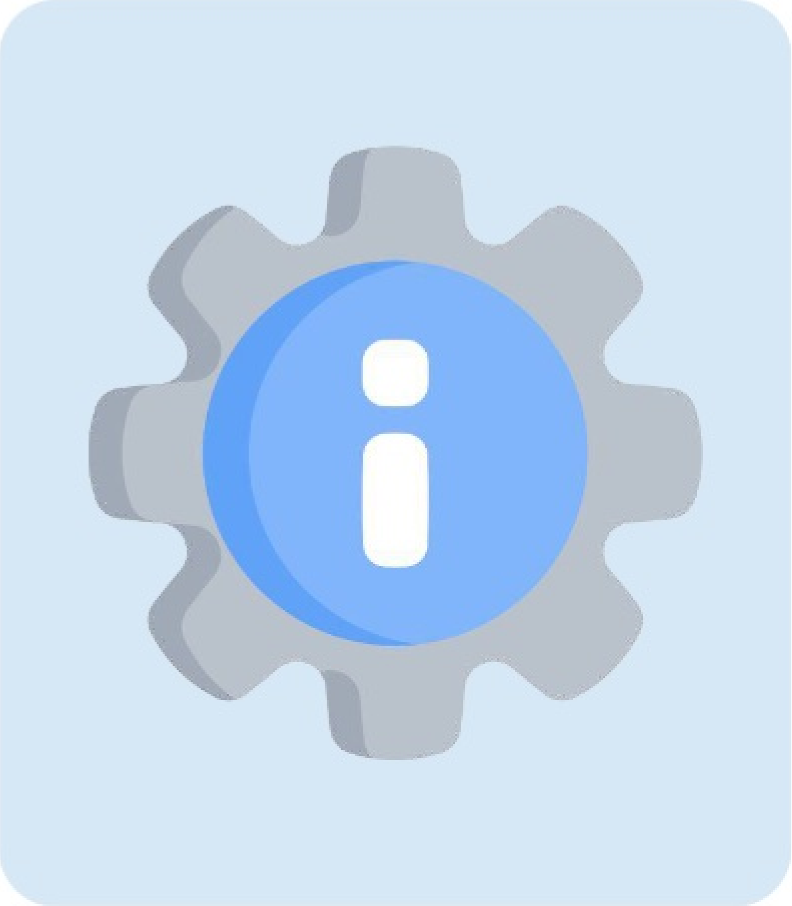
MEDIA TYPES

Unsupported media types can be mapped to these **supported media types** for request and response content serialization:

a) **application/json**
b) **application/x-www-form-urlencoded**
c) **text/plain**

SCHEMA

Certain considerations and limitations apply to schema definition. For instance, up to **10,000,000 characters (10 MB)** are supported. Additionally, methods that can be used in a schema are **GET, PATCH, PUT, POST, and DELETE**.



Apex REST Callouts

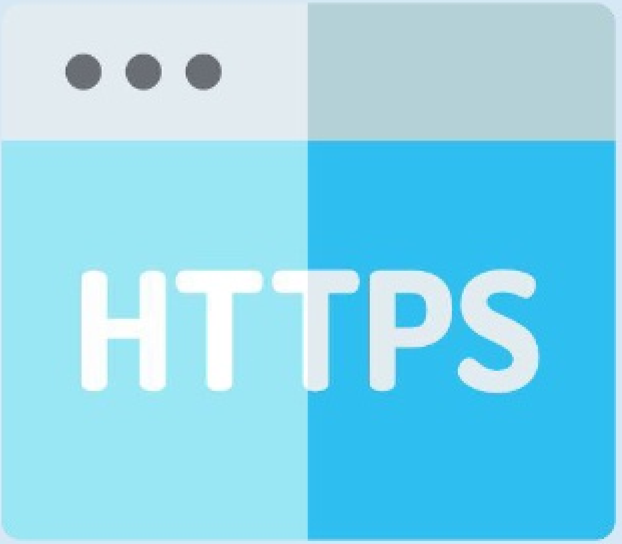
Certain considerations are applicable for using Apex code to perform an **HTTP/REST** callout to an external service for **retrieving, creating, updating or deleting** a resource.

REST CALLOUT ELEMENTS

A REST callout request requires an **HTTP method** and an **endpoint**. The following methods are supported: **GET, POST, PUT, DELETE, TRACE, CONNECT, HEAD**, and **OPTIONS**

HTTP CLASSES

The **Http** class is used to **initiate an HTTP request and response**. The **HttpRequest** class is used to **create an HTTP request**, such as a **GET, POST, PUT, or DELETE request**. The **HttpResponse** class is used to **handle the HTTP response** returned by the request.



Apex REST Callouts

Certain considerations are applicable for using Apex code to perform an **HTTP/REST** callout to an external service for **retrieving, creating, updating or deleting** a resource.

HTTP REQUEST ELEMENTS

The **HttpRequest** class has methods that can be used to set the **type of request**, **request headers**, **read timeouts and connection timeouts**, **redirects**, and **content of the message body**.

HTTP RESPONSE ELEMENTS

The **HttpResponse** class has methods that allow getting the **HTTP status code**, **response headers**, and **content of the response body**.

REMOTE SITE SETTING

A **remote site setting** must be created for the callout **endpoint**. This is a way of **authorizing** the **remote site** that hosts the external service. Alternatively, the callout can specify a **named credential** as the endpoint.

Apex SOAP Callouts

Apex code can be used to perform a **SOAP callout** to an external system. Certain considerations apply for **invoking an external SOAP web service** from Salesforce.

WSDL DOCUMENT

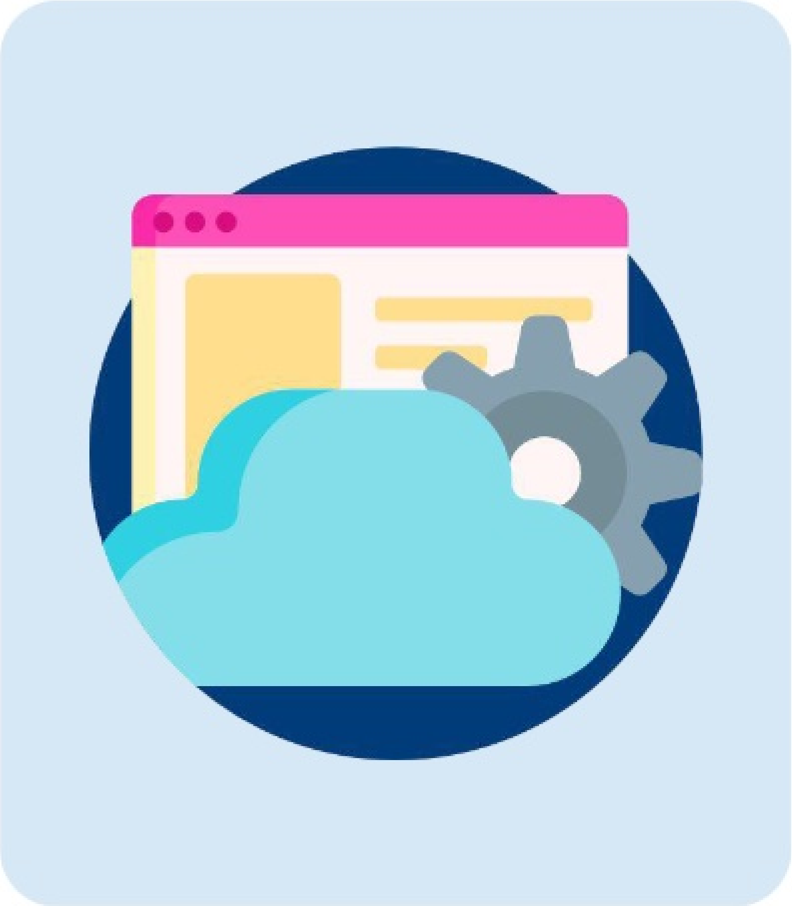
A **WSDL document** that represents the external web service must be obtained and stored on a local hard drive or network. This document allows making callouts.

APEX CLASSES

Apex classes can be generated by **consuming the WSDL document** on the **Apex Classes** page in Setup. Alternatively, the **WSDL2Apex** tool, which is open source and available on GitHub, can be used. The generated classes include **stub** and **type** classes for calling the external web service.

INVOCATION

To invoke the external service, an **instance of the stub** can be created and methods can be called on it in **Apex** code.



Apex SOAP Callouts

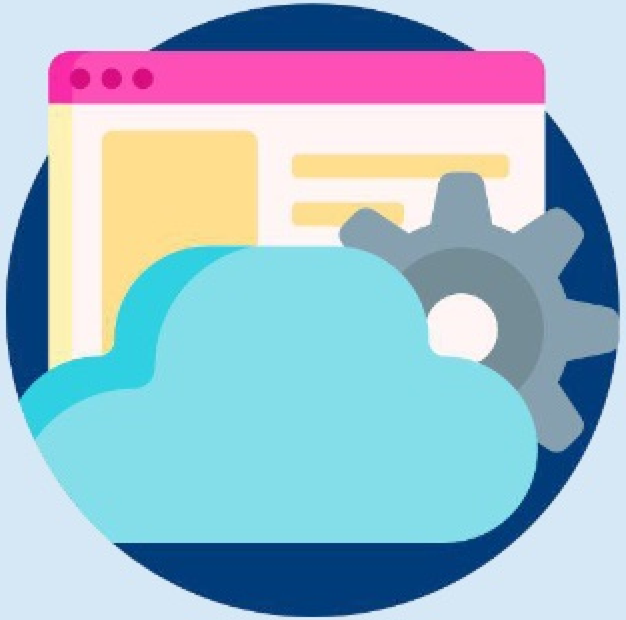
Apex code can be used to perform a **SOAP callout** to an external system. Certain considerations apply for **invoking an external SOAP web service** from Salesforce.

HTTP HEADERS

HTTP request and response headers can be set on a SOAP callout by adding **inputHttpHeaders_x** and **outputHttpHeaders_x** to the **stub**. For example, the value of a cookie can be set in the Authorization header.

REMOTE SITE SETTING

A **remote site setting** must be created to allow an **endpoint** to be accessed for the SOAP callout. Alternatively, the callout can specify a **named credential** as the endpoint.



Outbound Messaging

Certain considerations should be kept in mind for **sending outbound messages** to external systems.

NOTIFICATIONS

A **SOAP message** can include up to **100 notifications**. A notification contains the **object ID** and a **reference to the sObject data**.

USER PERMISSION

The user who is specified in the configured outbound message must have the **'Send Outbound Messages'** permission in their assigned profile or a permission set.

SESSION ID

While configuring an outbound message, **'Send Session ID'** must be selected to include a **session ID** if the listener should be able to make **API calls** back to Salesforce. The session ID represents the user specified in the outbound message.

Outbound Messaging

WSDL

The **WSDL** bound to the outbound message can be viewed on the **outbound message detail page**. It can be used to **generate stub listeners** in the external system.

CLIENT CERTIFICATES

The **endpoint server's SSL/TLS** can be configured to **require client certificates (two-way SSL/TLS)** to validate the identity of the Salesforce server.

STATUS

The **status of an outbound message** can be tracked on the **'Outbound Messages'** page in Setup.

Business Scenario

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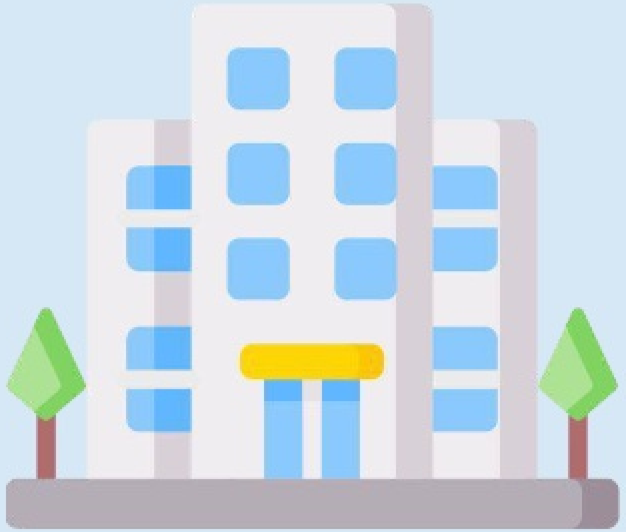
Business Scenario

The business scenario below should be reviewed to determine the most appropriate solutions for the company's requirements.



Cosmic Software Solutions is a company in the United States that specializes in software development. It offers its services to both individuals and businesses. Salesforce is mainly used by the company to manage customer accounts, contacts, and opportunities.

In addition, the internal users of the company use a project management system (PMS) and an employee management system (EMS). However, the company recently started using Salesforce for managing data related to certain types of projects and employees. Specific business processes that utilize the data in Salesforce need to be integrated with the other systems. The Salesforce architect of the company needs to design solutions for outbound integration requirements while identifying any related considerations.

Requirements & Solutions

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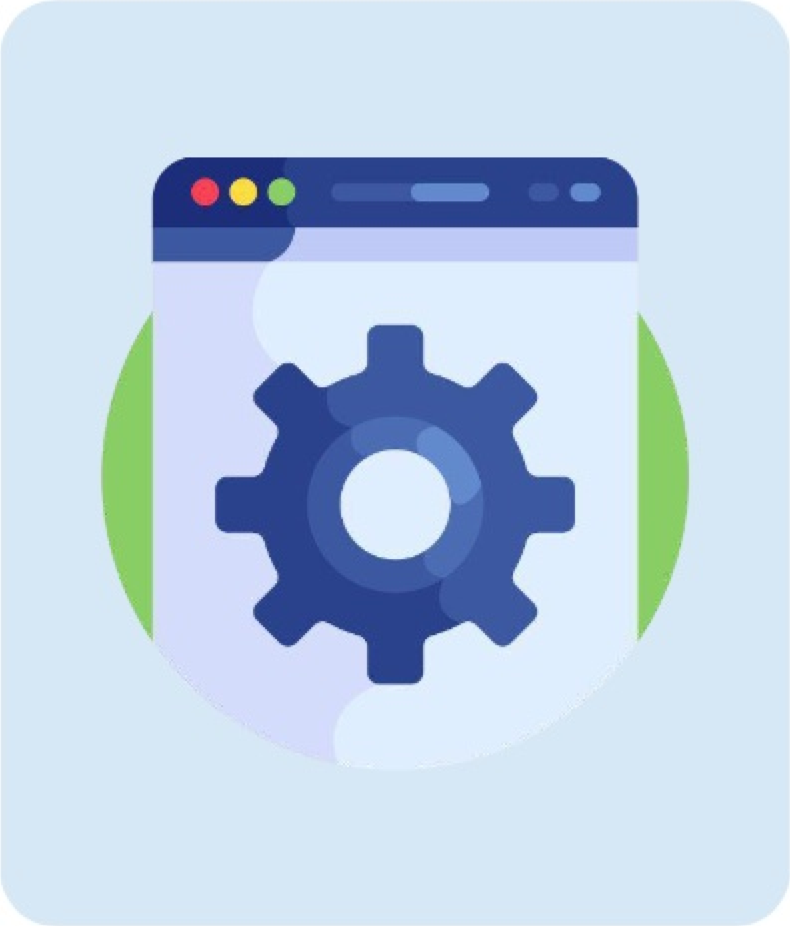
Requirement 1

SCENARIO

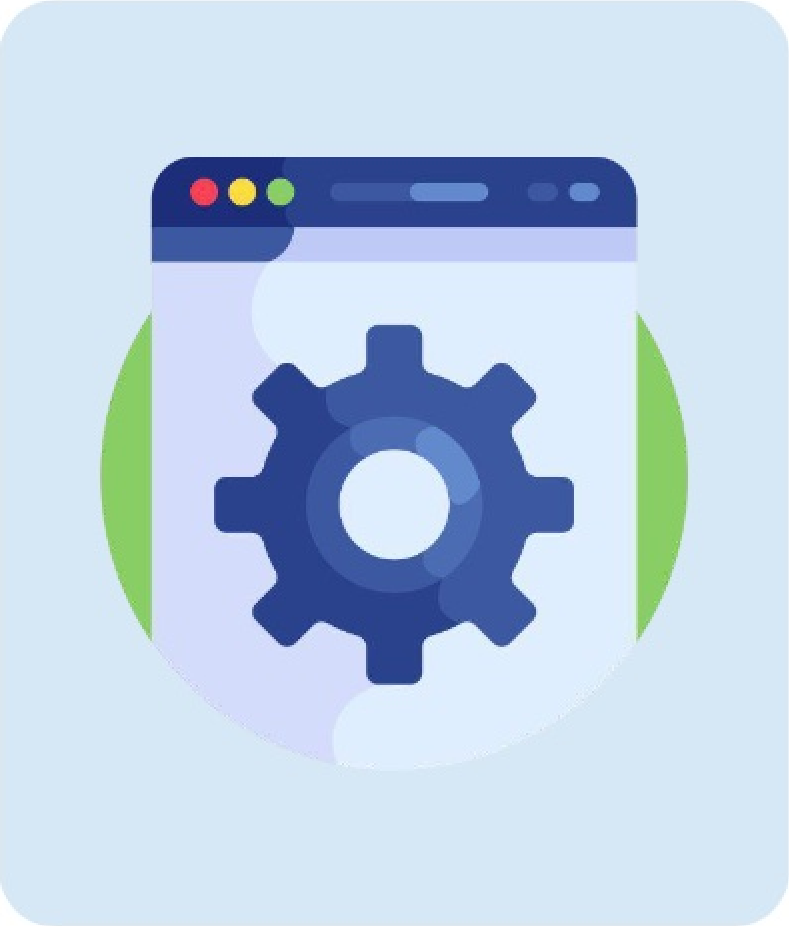
When a team of sales users needs to work on a new project, the assigned project supervisor creates a record of a custom object named ‘Project__c’ in Salesforce. The record is then created manually in the project management system (PMS). The sales director would like to automate the record creation by integrating the two systems. When a record is created in Salesforce, record details need to be sent to the PMS so that the corresponding record can be created in the target system automatically. The PMS has a RESTful service that can be used for record creation.



Solution 1

SOLUTION

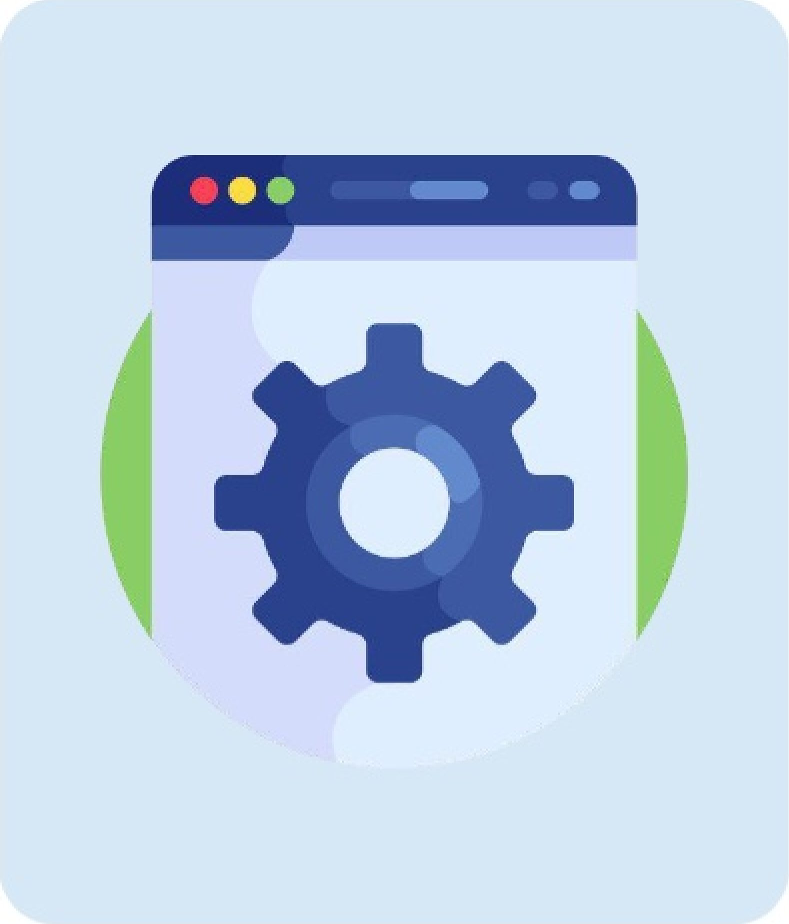
A **record-triggered flow** that invokes an **external service action** can be used to meet this requirement. The action can create a record in the PMS after the original record in Salesforce is committed. However, it is important to consider that the **API specification** of the RESTful service in the PMS must be available in **OpenAPI 2.0** or **OpenAPI 3.0 JSON schema** format. It should include the operation required for creating a new project record. It is also necessary to define a **named credential** for authenticating the external service, which must be specified while defining the external service.



Requirement 2

SCENARIO

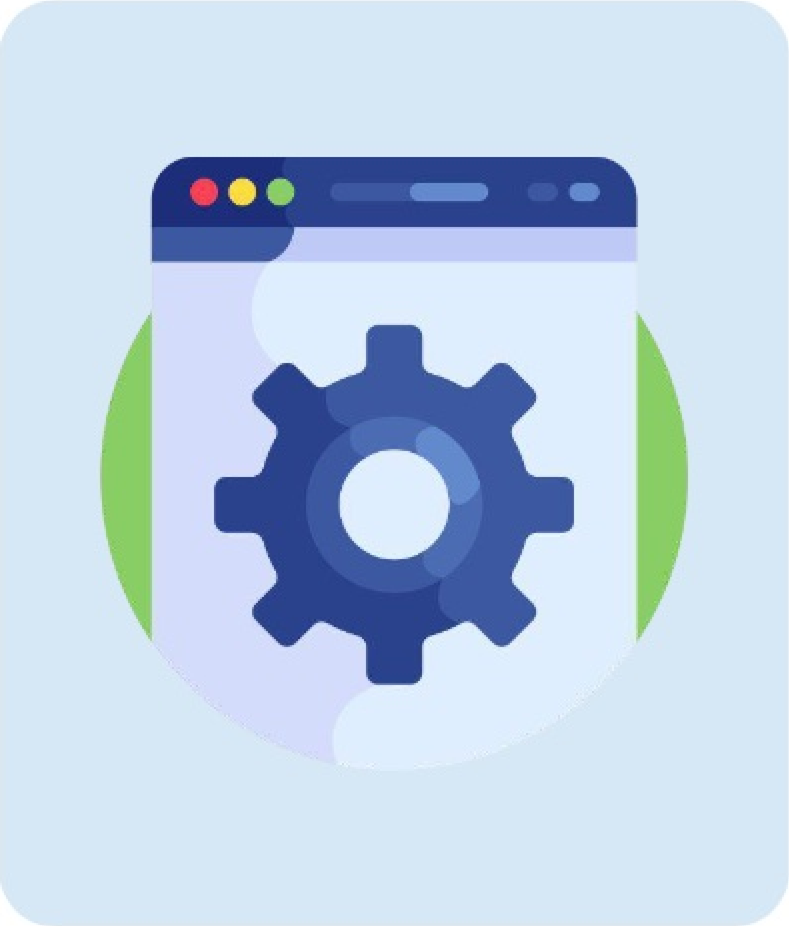
A custom object named 'Employee__c' has been created in Salesforce to store information about sales users. When a sales user is entitled to extra compensation for additional services, an HR specialist should be able to use a button on the user's employee record in Salesforce to send related information to the employee management system (EMS). The EMS should then update the compensation details of the employee. It has a SOAP web service that can process the received data for this use case.



Solution 2

SOLUTION

A custom **quick action** that invokes a **flow** can be used for this requirement. The flow can perform an **Apex SOAP callout** to the SOAP web service hosted by the EMS. However, the **WSDL document** of the web service must be obtained so that **stub and type Apex classes** can be generated by consuming it on the **Apex Classes** page in Setup or using the **WSDL2Apex** utility. A **remote site setting** must be created for the endpoint of the web service in Salesforce to authorize it for the SOAP callout. To invoke the web service, an **instance of the stub** can be created and methods can be called on it in **Apex**.



Requirement 3

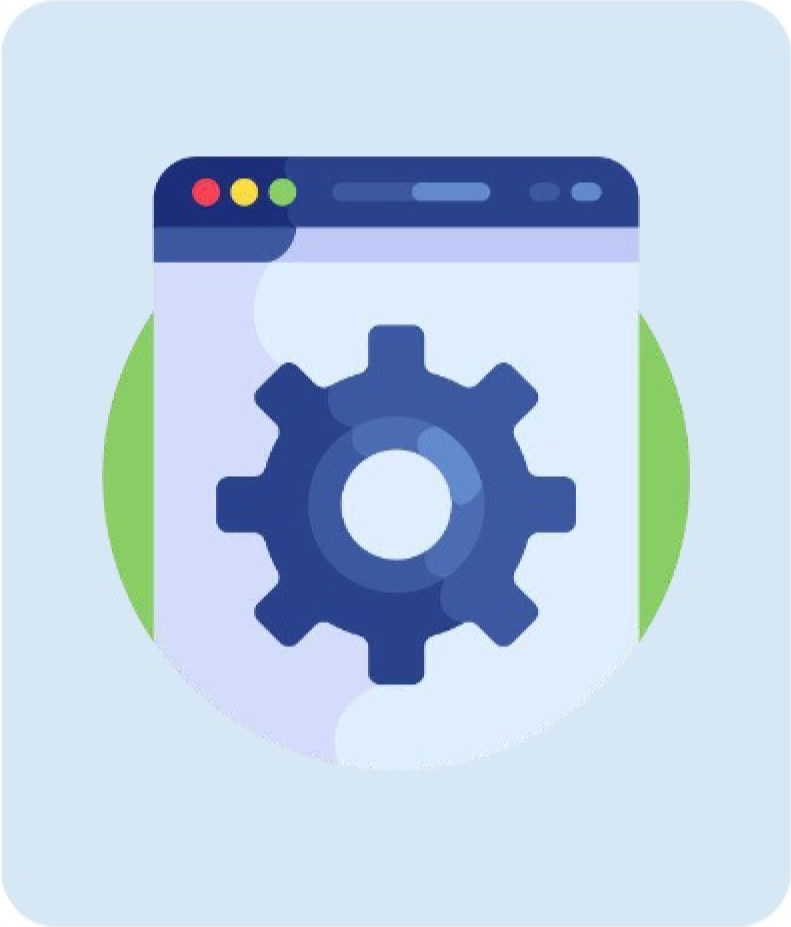
SCENARIO

A RESTful service has been created in the employee management system to allow a system administrator to perform operations such as getting a list of employee records, creating a new employee record, and updating the information of an existing employee. A system administrator can use a third-party application such as Postman to perform these operations, but the HR director would like to make them available to HR managers who can access Salesforce. A new user interface is required for this purpose.

Solution 3

SOLUTION

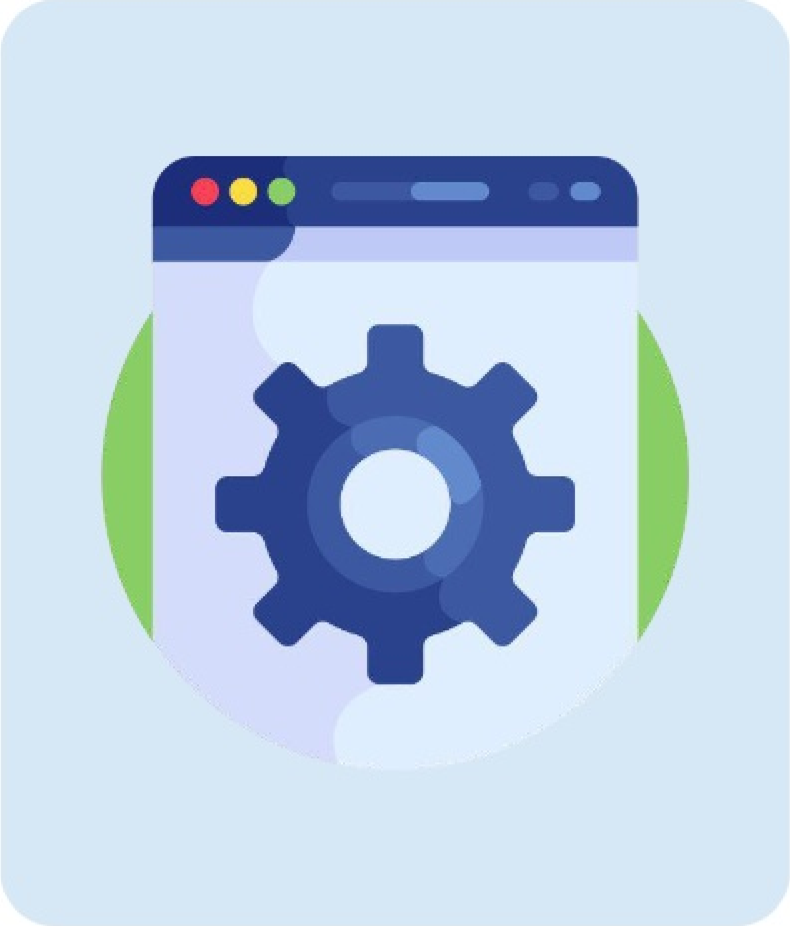
A custom **Lightning web component** can be created for this requirement. It can contain buttons associated with methods in an Apex controller for performing different types of **HTTP callouts**. A **remote site setting** must be created for the endpoint of the RESTful service. The **GET, POST, and PUT methods** can be used in the Apex code to get a list of employees, create a new employee, and update an existing employee, respectively. The Http class provides the capability to initiate HTTP requests and receive responses. The **HttpRequest** class can be used to create a new HTTP request. The **HttpResponse** class can be used to handle the response received by the RESTful service.



Requirement 4

SCENARIO

A project team can consist of several members. When a new team member is added to a project, information about their role and responsibilities is added to a 'Text Area (Long)' field on the project record in Salesforce. This information needs to be sent to the project management system (PMS) automatically so that it can be stored on the corresponding project record. A SOAP endpoint can be configured in the PMS for this use case.



Solution 4

SOLUTION

An **outbound message** action invoked by a **record-triggered flow** can be utilized for this requirement. The outbound message can be sent as a specific Salesforce user. It is also possible to send the **session ID** to enable the endpoint to make callbacks to Salesforce. Once the outbound message is defined, its **WSDL** must be downloaded. The WSDL defines the messages that Salesforce sends to the SOAP endpoint and can be used to generate **stub listeners**. The **status** of a sent outbound message can be tracked on the **'Outbound Messages'** page in Setup.

Requirement 5

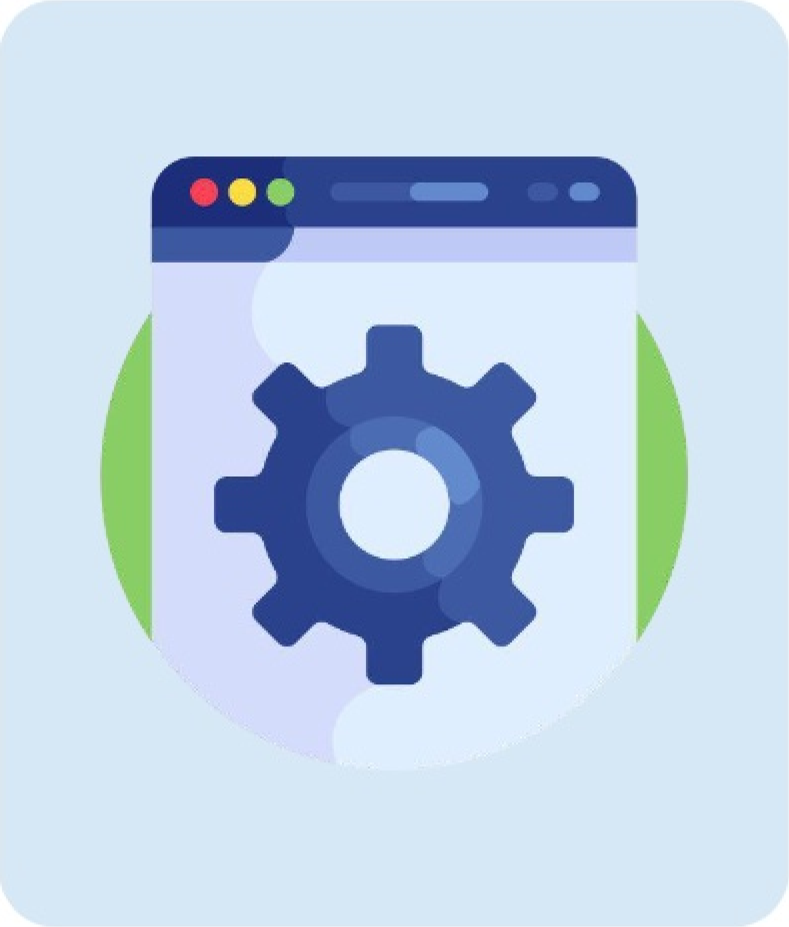
SCENARIO

The company has built an Experience Cloud site that can be accessed by its employees. The site uses an Experience Builder template. The employee management system is used by some internal users to manage sensitive employee data. The company would like to integrate the Experience Cloud site with the external system declaratively to allow these users to update employee details stored in the external system.

Solution 5

SOLUTION

External Services and a **Flow** can be utilized for the given requirement. The **Flow** component can be added to an Experience Cloud site page. A flow selected in the component can invoke an **external service** action that updates an employee record in the employee management system based on the data entered by a user. It is important to note that Experience Builder sites **do not support using iframes**, so they cannot be used for the requirement. Also, invoking Apex code or utilizing a Lightning web component would not meet the requirement declaratively.



Learn More

- Get Started with External Services
- Understanding Outbound Messaging
- Apex REST Callouts
- Apex SOAP Callouts